

Quick Activity 2: Canteen Group Order

A canteen sells **rice meals at ₱85 each** and **drinks at ₱25 each**. Every order receives a **10% discount on the rice-meal subtotal only**. The canteen also charges a **₱5 packaging fee per rice meal**.

Problem Analysis

Input	Process	Output
Number of rice meals, number of drinks, payment amount	Calculate rice meals Subtotal, calculate the 10% of rice meal Subtotal, subtract the discount, calculate drink total, add the packaging fee per rice meal, calculate change by subtracting the final bill from the payment	mealSbttotal, Drink subtotal, Discount, Packaging fee, Final bill, Change

Decomposition

1. Input the number of rice meals.
2. Input the number of drinks.
3. Input the payment amount.
4. Calculate the meal subtotal.
5. Calculate the drink subtotal.
6. Calculate the 10% discount on the meal subtotal.
7. Calculate the packaging fee.
8. Calculate the final bill.
9. Calculate the change.
10. Display all monetary results to two decimal places.

Pseudocode

START

```
READ ricemealSubtotal drinkCost, discountedSubtotal, packagingFeePerMeal, payment
ricemealSubtotal ← quantity × 85
discount ← ricemealSubtotal × 10%
discountedSubtotal ← ricemealSubtotal – discount
drinkCost ← quantity × 25
packagingFeePerMeal ← quantity × 5
totalCost ← discountedSubtotal + drinkCost + packagingFeePerMeal
change ← payment – totalCost
DISPLAY totalCost, change
```

END

Dry Run Calculations

Test Case 1

Input

- Rice Meals = 3
- Drinks = 4
- Payment = ₱500.00

Step	Variable	Calculation	Result
1	Rice mealSubtotal	$3 \times \text{₱}85.00$	₱255.0
2	Discount	$0.10 \times \text{₱}255.00$	₱25.50
3	discountedSubtotal	$\text{₱}255 - \text{₱}25.50$	₱229.50
4	drinkCost	$4 \times \text{₱}25.00$	₱100.00
5	packagingFeePerMe	$3 \times \text{₱}5.00$	₱15.00
6	totalCost	$\text{₱}229.50 + \text{₱}100.00 + \text{₱}15.00$	₱344.50
7	Change	$\text{₱}500.00 - \text{₱}344.50$	₱155.50

Test Case 2

Input

- Rice Meals = 5
- Drink = 3
- Payment = ₱600.00

Step	Variable	Calculation	Result
1	Rice mealSubtotal	$5 \times \text{P}85.00$	₱425.00
2	Discount	$0.10 \times \text{₱425.00}$ ₱42.50	₱75.00
3	discountedSubtotal	₱425.00 - ₱42.50	₱42.50
4	drinkCost	$3 \times \text{P}25.0$	₱25.00
5	packagingFeePerMe	$5 \times \text{P}5.0$	₱482.50
6	totalCost	₱382.50 + ₱75.00 + ₱25.0	₱482.50
7	Change	₱600.00 - ₱482.50	₱117.50

Explain: Why would applying the 10% discount to the entire bill produce an incorrect result? According to the instructions, the 10% discount should be applied only to the Rice Meals subtotal. If it is applied to the entire bill, the drinks and packaging fee will also receive a discount, which goes against the given rules. This will result in an incorrect final bill.